Chris Harper

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Staff AI Enablement Engineer | Agentic Engineering | Developer Tooling | Engineering Leadership

SUMMARY

Staff AI Enablement Engineer and former engineering manager with 18 years building web and cloud platforms. Built the AI-assisted engineering platform the Salient Systems web team now works in on Claude Code: custom skills, guardrail hooks, review and research agents, and cost governance. Earlier, helped rebuild Salient's CompleteView video web client, architected HIPAA-compliant AWS platforms for ALSAC/St. Jude, and led consulting teams for clients including Capital One, Rivian and GitKraken. Ships products independently, including a real-time neural amp modeling app for Android and an autonomous AI news pipeline.

SKILLS

- **AI Engineering:** Claude Code (skills, hooks, subagents, MCP) • LLM agent orchestration • RAG • pgvector • Vectorize • Workers AI • QLoRA fine-tuning • OpenAI API • AWS Bedrock • LangChain
- **Languages & Frameworks:** TypeScript • Python • C++ • Angular • React • React Native • Next.js • Node.js
- **Platforms & Infrastructure:** AWS (Lambda, ECS, API Gateway, DynamoDB) • Cloudflare Workers • Supabase • PostgreSQL • MongoDB • Docker • Kubernetes • OpenTofu • GitHub Actions • Azure • GCP
- **Video & Performance:** WebCodecs • Web Workers • WebSockets • H.264/H.265 • Chrome DevTools Protocol profiling
- **Leadership:** Engineering management • Technical strategy • Cross-team planning • Hiring and mentoring • Agile

PROFESSIONAL EXPERIENCE

Staff AI Enablement Engineer | Salient Systems

Oct 2025 – Present • Promoted from Senior Software Engineer, Nov 2026

Grew the role from fixing the legacy CompleteView video surveillance web client to rebuilding it, then to building the AI-assisted engineering practice the web team now works in.

- Designed and built the web team's AI engineering platform on Claude Code, authoring 30 of its 31 custom skills plus most of its 27 hooks, 8 agents and 41 rule files (102 of 136 tooling commits)
- Automated ticket delivery end to end: verified work lists, test-first fixes, browser verification, parallel review agents and Jira evidence for QA, behind hooks that gate every push and Jira transition
- Set the team's agent cost governance from measured multi-agent runs, built a sessions dashboard for parallel AI sessions, and built an AI-driven course generator that outputs Rise and SCORM 1.2 packages
- Wrote the System Modernization Strategy for executive leadership, then helped rebuild the web client as an Angular 22 TypeScript monorepo, becoming its top committer (186 merged PRs, 161 Jira tickets closed)
- Built the browser video player for H.264, H.265 and MJPEG on WebCodecs and Web Workers, and shipped AI-detection Enhanced Search and the Alarm View operator console
- Planned company-wide single sign-on across web, mobile, cloud and backend teams (73 tickets, 10 epics) and reviewed or commented on 236 teammate PRs

Cloud Architecture Consultant (Contract) | ALSAC/St. Jude Children's Research Hospital

May 2023 – Aug 2025

- Architected solutions for an enterprise healthcare platform handling 10M+ daily messages under HIPAA requirements
- Designed real-time microservices on Python, AWS Lambda, ECS and API Gateway
- Rolled out Infrastructure as Code across 15+ development teams, cutting environment provisioning from days to minutes

- Built CI/CD pipelines with integrated security scanning (99.5% deployment success) and reached sub-100ms API responses while cutting AWS costs 30%

Engineering Manager & Technical Lead | Blue Sky Cloud Consulting
Aug 2020 – Apr 2023

- Managed full-stack consulting engagements for clients including GitKraken and Rivian
- Architected enterprise SaaS platforms with real-time messaging, workflow automation and analytics for 10,000+ active users
- Built real-time telemetry and predictive analytics for manufacturing IoT devices
- Led technical hiring and mentorship, with 95% team retention and an 80% internal promotion rate

Senior Software Engineer & Team Lead | Cerity
Aug 2018 – Aug 2020

- Led a full-stack team building a new insurance platform on Angular, React, Python and serverless AWS
- Designed serverless applications supporting 50,000+ concurrent users with auto-scaling Lambda functions

Lead Software Engineer | ThunderDork LLC
May 2015 – Aug 2018

- Led software development for Fortune 500 clients including Capital One, managing distributed teams
- Designed a microservices migration on Docker and AWS that let teams deploy independently
- Set testing frameworks and development standards that cut production incidents 40%

Senior Engineering Roles | Netspend • Cinsay • Schematic
2008 – 2015

Built financial platforms serving millions of users, video-based shopping experiences, and award-winning interactive applications for Fortune 500 brands including Dell.

INDEPENDENT PROJECTS

2026 – Present, alongside full-time role, built with Claude Code

- **AmpHoarder (amphoarder.com):** Building a free, open-source Android guitar rig in React Native and C++ that plays Neural Amp Modeler captures live at 7.08 ms round-trip latency, with 24 built-in effects and MIDI foot control, now in Google Play closed testing
- **cloudcodetree.com:** Built and run a Next.js site on a Cloudflare Worker where a scheduled Claude agent publishes AI news three times a day (951 articles), with hybrid keyword and semantic search, Supabase auth with row-level security, and infrastructure in OpenTofu
- **AI Engineering Curriculum:** Published 11 hands-on tutorials on RAG, fine-tuning and Claude Code, and writing DealFinder, a 37-part AI engineering course with a companion app (237 tests) covering QLoRA, ReAct agents, MCP, pgvector RAG, MLOps and Kubernetes

EDUCATION, CERTIFICATIONS & AWARDS

- **Associate of Applied Arts, Computer Animation** | The Art Institute of Dallas
- **AWS Certified Developer – Associate** • Multi-Cloud Platform Certifications (Azure, GCP)
- **Webby People's Choice Award Winner** – Digital Innovation Excellence